

Practice Quiz: Cognition

Part A: Multiple Choice

1. Precision in Active Inference is mathematically defined as: A) The mean of a probability distribution B) The inverse variance ($1/\sigma^2$) — a measure of confidence or certainty C) The entropy of a distribution D) The log-likelihood ratio
2. Neurally, precision is proposed to be implemented by: A) The firing rate of inhibitory interneurons B) The synaptic gain of superficial pyramidal cells carrying prediction errors C) The myelination thickness of axons D) The density of dendritic spines
3. Acetylcholine primarily modulates: A) Precision on top-down predictions B) Precision on bottom-up (sensory) prediction errors C) Motor neuron firing rates D) Sleep-wake transitions
4. The ADHD phenotype in Active Inference may reflect: A) Excessive sensory precision leading to constant distraction B) Low tonic dopamine → difficulty sustaining precision on task-relevant prediction errors C) Damage to the prefrontal cortex D) An excess of acetylcholine
5. Lawson et al.'s (2014) account of autism proposes: A) Low sensory precision leading to reduced perception B) High precision on sensory prediction errors, reducing the influence of top-down contextual priors C) Normal precision with impaired motor control D) An emotional deficit unrelated to perception
6. Anxiety in the Active Inference framework is characterized by: A) No prediction errors at all — the anxious person predicts perfectly B) Excessive precision on threat-related prediction errors, biasing the generative model toward danger detection C) Absence of interoceptive awareness D) Overactivity exclusively in the amygdala
7. The neuromodulator most associated with coding environmental volatility (how changeable the world is) is: A) Dopamine B) Serotonin C) Noradrenaline D) Acetylcholine

Part B: Short Answer

1. Explain the "cocktail party effect" (attending to one voice in a noisy room) in Active Inference terms. Which neuromodulatory system is most likely involved?
2. Patient with autism reports extreme sensitivity to fluorescent lighting but excels at detecting visual anomalies in industrial quality control. Analyze both the challenge and the strength using the precision framework.
3. An SSRI (selective serotonin reuptake inhibitor) reduces anxiety symptoms within 4-6 weeks. Propose a mechanism of action in Active Inference terms: how might increased serotonin availability reduce the excessive precision on threat-related prediction errors?